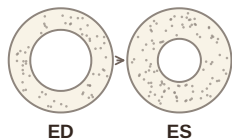


a Estimation track: cine supplies cues; correspondence is inferred; strain is defined by convention

● 1 · OBSERVED

Cine bSSFP, short axis



borders conspicuous;
intramural texture weak;
no material label

①

● 2-3 · INFERRED

Estimated displacement $u(X, t)$



the data term D does not
identify the map;
the prior αR selects:
tracking · registration ·
biomechanics · learned

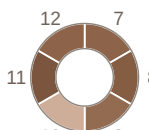
$$\hat{\varphi} = \arg \min D(I_0, I_t \circ \varphi) + \alpha R(\varphi)$$

identical masks admit different material
maps; the prior selects among them
(Fig. 2)

Ⓔ Ⓓ

● 4 · DEFINED

Green-Lagrange Ecc, pixelwise



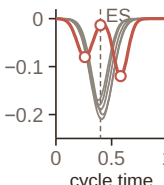
fill = Ecc; seg 11 reduced

choices 01-05 fix the value before any
region is reported

- 01 ED reference
- 02 $E = \frac{1}{2}(F^T F - I)$
- 03 c/r/l axes, signed
- 04 endo-mid-epi, 2D
- 05 $\partial u / \partial X$, smoothed

● 5 · REPORTED

Segmental Ecc(t), mid ventricle



- 06 AHA mid 7-12
- 07 at ES **-0.01**
- peak sys. **-0.08**
- post-sys. **-0.12**
- 08 min(mean) **-0.16**
- mean(min) **-0.18**

choices 06-08 pick one number from the
same curves (○ segment 11)

Failure points, each testable alone:

① input sufficiency · re-image the motion more finely

Ⓔ estimation · vary α with images fixed

Ⓓ representation · project the known field onto the basis

b Validation track: each reference supports one claim; results as reported

● DENSE ● tagging ● cine FT ● cine DL ○ comparator

Known motion

MRXCAT2.0 test of
DeepStrain

- ✓ capacity, technical error
- ✗ not alone: in-vivo
accuracy

estimator error (mean $\pm$ SD)



○ infarct case, EF 49%;
prescribed scar Err
0.95 $\rightarrow$ 0.30 (Fig. 4)

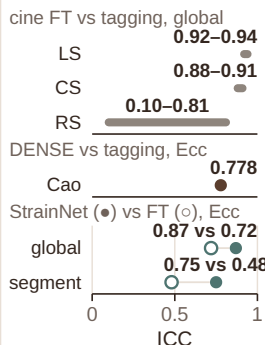
Also Dice 0.82; displacement
1.0 $\pm$ 0.9 mm. Slice means
only: no scar-region error.

Paired DENSE / tagging

Militaru 2021; Cao 2018;
Wang 2023

- ✓ material-sensitive
agreement
- ✗ not alone: reference error

ICC; range = 3 vendors



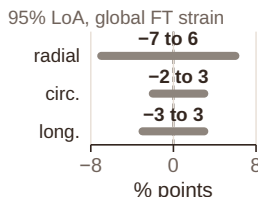
Militaru $n = 61$, 3 T; Cao,
DENSE vs tagging; StrainNet
(DL), 59 cine tests, DENSE
reference.

Precision

Yoon 2023; Bell 2025

- ✓ precision at tested scale
- ✗ not alone: accuracy (bias)

REGAIN vs GRAPPA



Strain-8 scan-rescan

same day, 20 volunteers $\times$ 8 protocols

global **fair-excellent**
segmental **more variable**

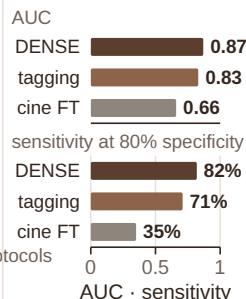
Mean global differences near
zero; radial limits widest (49
participants).

Tissue (LGE)

Kihlberg 2020

- ✓ scar-associated
dysfunction
- ✗ not alone: motion error

segments with LGE > 50%



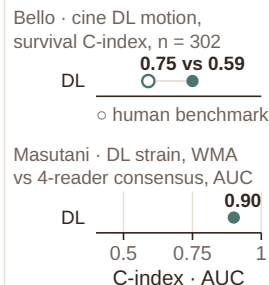
116 suspected CAD, 34 with
qualifying scar. Ranks scar
sensitivity, not motion
accuracy.

Outcome & readers

Bello 2019; Masutani 2023

- ✓ prognostic / diagnostic link
- ✗ not alone: clinical utility

discrimination, chance 0.5



Masutani: 53 patients, 846
segments, 30% peak Err
cut-off. C-index and AUC are
not comparable.